Operations

Week 4 – Live Class

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THREE MEASURES OF PROCESS PERFORMANCE

Cycle Time (time/unit)

- Average time between the output of successive units (e.g. 5 min/unit)
- **Flow rate** or **throughput rate**) is the inverse of the cycle time (e.g. 1/5 units /min = 0.2 units/min)

Flow time or manufacturing lead time or service lead time

- The average amount of time it takes a unit to get through the process (includes waiting time).

Inventory or Work in process (WIP)

- Average number of units in the process at any point in time (includes the units waiting to be processed).

Reasons to hold inventory

Seasonal inventory	• To absorb seasonal variations in supply or demand
Cycle inventory	• Economies of scale (Batching)
Decoupling inventory	• To absorb variations in the production rates of different stages
Safety inventory	• To hedge against the uncertainties in supply or demand
Pipeline inventory	• Production and distribution are not instantaneous

Tradeoffs

Seasonal Stocks

- Capacity vs inventory costs

Cycle Stocks

- Order or setup costs versus inventory costs

Safety Stocks

- Stockouts costs vs inventory costs

Setups Times

Setups: non-value added activities required to begin a process.

Setups result in interruptions to the process flow.

Ideally, we would like either to:

- Eliminate setups times
- Try to reduce the time it takes to perform the setup

Discussion: Impact of setup times in day-to-day life or at work

- How do setup times impact your day-to-day life? (e.g. supermarket example)

- Think about examples of setup times at work. How do they affect productivity? How do you deal with them?

How does setup impact capacity? – An example

Consider a milling machine which produces parts for a scooter:

- **Steer support (1 per output unit (scooter))**
- **Ribs (2 per output unit (scooter))**

Activity times and setup times

- 1 minute = Time to produce one steer support unit (of which there is one per scooter).
- 60 minutes = Time to change over the milling machine from producing steer supports to producing ribs (setup time).
- 0.5 minute = Time to produce a rib; since there are two ribs in a scooter, this translates into 1 minute/unit
- 60 minute = changeover time to bring it back to producing steer supports

Different batch sizes and their impact on capacity

Note that the capacity can be increased by increasing the batch size.

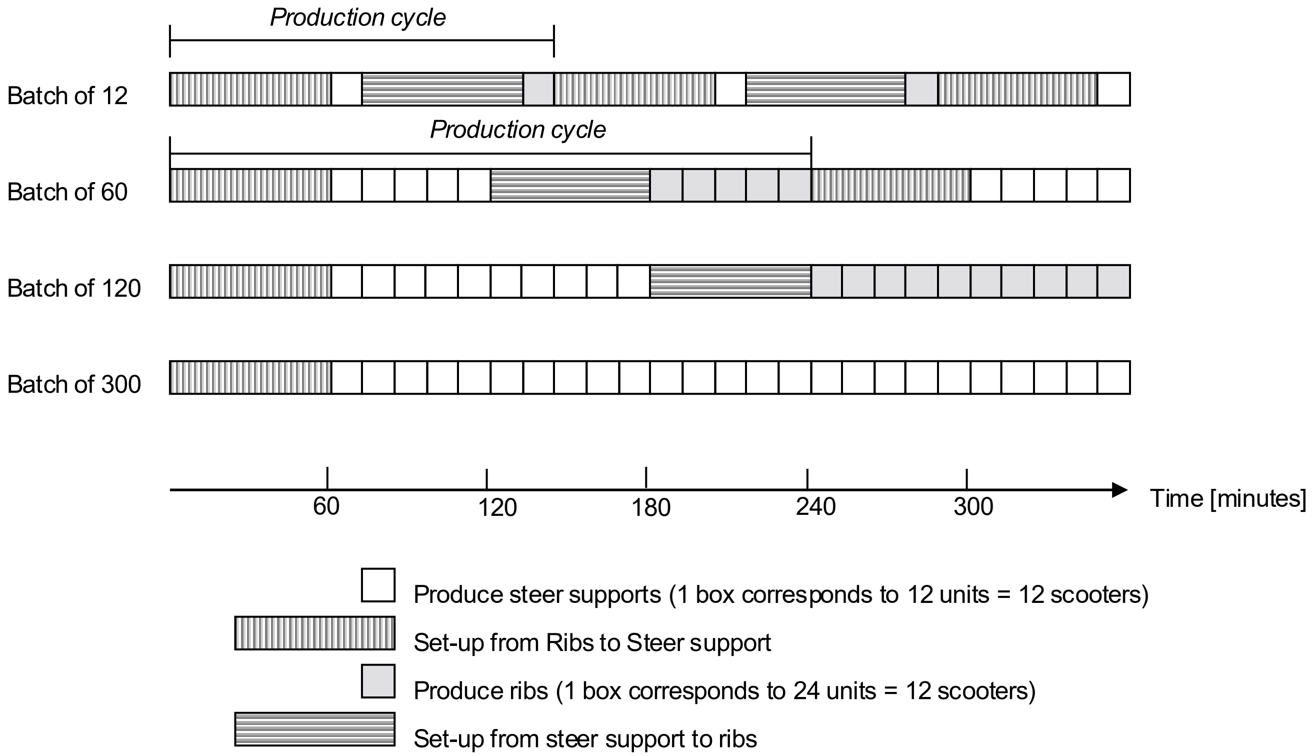

Figure 6.2: The impact of Set-up times on Capacity

Impact of setup on capacity

Batch Size	Time to complete one batch (minutes)	Capacity (units/minute)
12	60+12+60+12=144	12/144 = 0.0833
60	60+60+60+60 = 240	60/240 = 0.25
120	60+120+60+120 = 360	120/360 = 0.33
300	60+300+60+300 = 720	300/720=0.4166

$$\text{Capacity given Batch Size} = \frac{\textit{Batch Size}}{\textit{Setup time + Batch size*time/unit}}$$

Process Analysis with Batching

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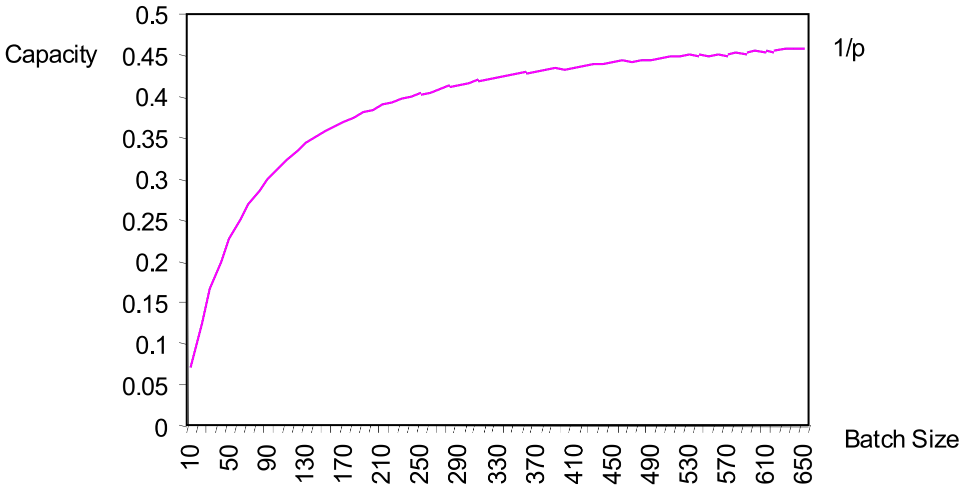
- Capacity calculation:

$$\text{Capacity given Batch Size} = \frac{\textit{Batch Size}}{\textit{Set-up time + Batch size*Time per unit}}$$

- Note: Capacity increases with batch size:



Choosing a batch size in the presence of setup time

Two key strategies

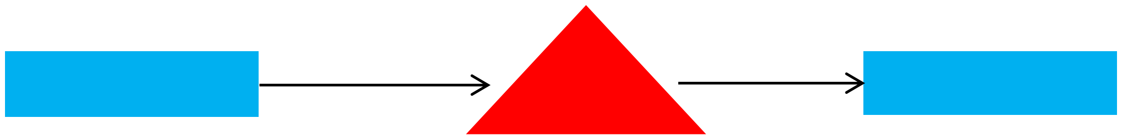
Case 1 – Setup occurs at the bottleneck activity

- It is desirable to increase the batch size
- Increase the batch size up to when another resource becomes the bottleneck.

Case 2 – Setup occurs at a non-bottleneck activity.

- It is desirable to decrease the batch size
- Inventory is reduced
- Decrease the batch size as much as possible without reducing the process capacity.

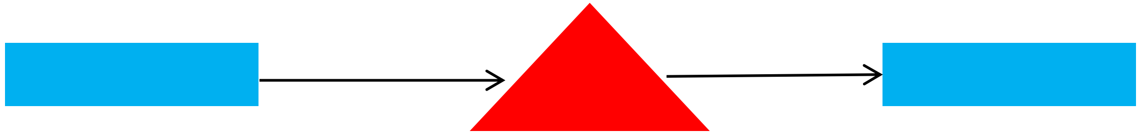
Example



Milling Machine
Setup time = 120 min
Per unit time = 2 min/unit

Assembly Process
No setup time
Cycle time: 3 min / unit

Example



Milling Machine
Setup time = 120 min
Per unit time = 2 min/unit

Assembly Process
No setup time
Cycle time: 3 min / unit

Capacity of the assembly process ~= 0.33units/min

Case 1: Batch size = 12

- Capacity Milling machine: 12 units / (120 min + 12 x 2 min) ~= 0.0833 units/min (**Bottleneck**)

Case 2: Batch size = 300

- Capacity Milling machine: 300 units / (120 min + 300 x 2 min) ~= 0.4166 units/min (**Non-Bottleneck**)

Recommended batch size

We are looking for the batch size that leads to the lowest level of inventory without affecting flow rate:

$$\text{Flow rate} = \frac{\textit{Batch Size}}{\textit{Setup time} + \textit{Time per unit} \times \textit{Batch size}}$$

Rearranging the terms, we get

$$\text{Recommended Batch Size} = \frac{\textit{Flow rate} \times \textit{setup time}}{1 - \textit{Flow rate} \times \textit{time per unit}}$$

Practice question 1 (past challenge)

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We consider a single processing step that produces two types of units: A and B for the production of solar panels.

- Each solar panel requires one unit of A and one unit of B.

The process step needs some time to switch from producing A to B as well as from B to A.

- **Setup time to switch from A to B:** 30 minutes.
- **Setup time to switch from B to A:** 45 minutes.
- The speed at which it can produce type A units is 1 unit per minute. This means it has a processing time of one minute per unit.
- The speed at which it can produce type B units is 0.5 units per minute. This means it has a processing time of two minutes per unit.

The firm decided to produce solar panels 24 hours per day, seven days a week, using a production cycle with a batch size of 25 solar power units (ie: 25 As and 25 Bs).

This means that firm will be producing 25 units of type A, then 25 units of type B, then 25 units of type A, then 25 units of type B,etc,etc,etc.

Calculate the capacity of this process given this batch size.

Solution to practice question 1

Time taken to run a complete batch cycle:

$$25 \times 1\ min + 25 \times 2\ min + 30\ min + 45\ min = 150\ min.$$

Capacity: $\frac{25}{150} \simeq 0.16666\ solar\ panels\ per\ min = 10\ solar\ panels\ per\ hour.$

Practice question 2 (past challenge)

Suppose the firm mentioned in Question 1 wants to have a process capacity of 18 solar power units per **hour**.

What should the batch size be to accomplish this?

*State your answer as a recommended **batch size**.*

Solution to practice question 2

Explanation

We can use the recommended batch size formula:

Desired flow rate: $18\ per\ hour = \frac{18}{60} = 0.3\ units\ per\ minute$

Total setup time per batch: $75\ minutes$

Total time per unit: $1\ min + 2\ min = 3\ min$

$$Recommended\ batch\ size = \frac{0.3 \times 75\ min}{(1 - 0.3 \times 3)} = 225.$$

Inventory model with setup costs

Main model to understand setup costs:

Economic Order Quantity model (EOQ).

Possible scenario: A retailer needs to schedule purchases from the supplier to satisfy demand.

Economic Order Quantity (EOQ)

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Assumptions

- Demand is constant (i.e. flow rate is constant)
- Fixed cost per order
- Purchase price is constant
- Instantaneous delivery

Decision variables

- When to order?
- How much to order?

Economic Order Quantity

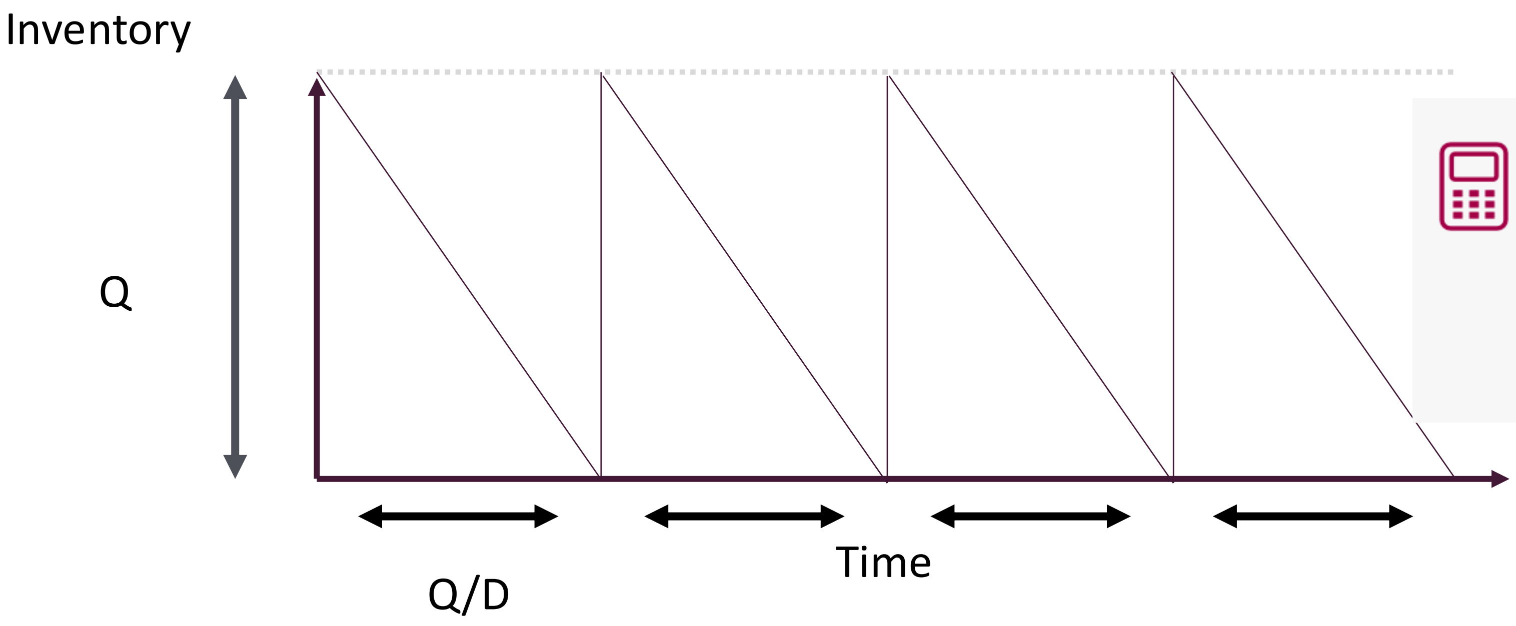
What do we need

- Order cost: Co (fixed) (e.g. shipment)
- Demand rate: *D (output units/ unit of time*))
- Inventory (holding) cost: Ch ($/per unit of time and unit of product)

What do we want to know

- Optimal order quantity Q to minimise inventory and ordering costs.
- Schedule of the orders.

Inventory and Ordering Costs



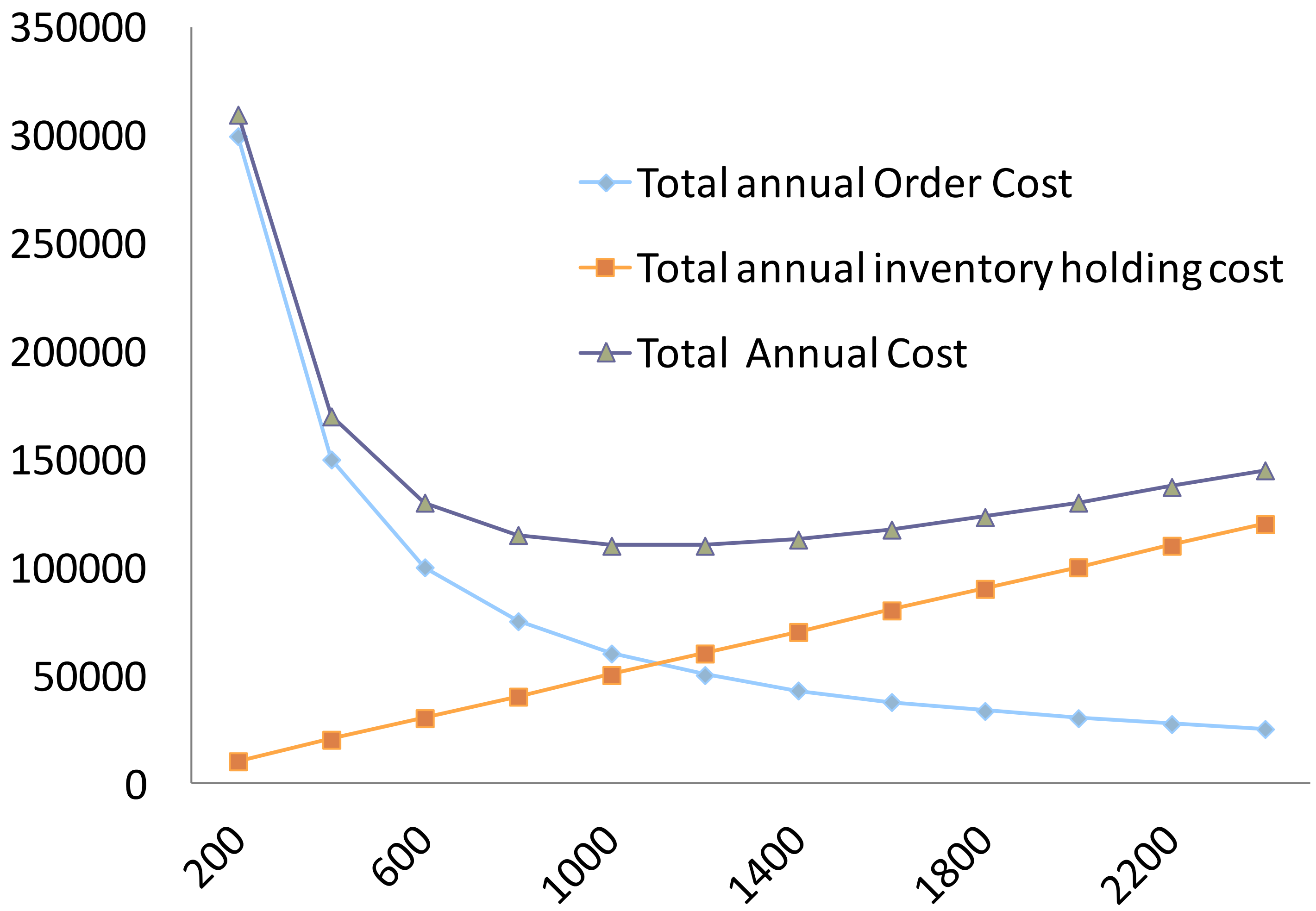
Average ordering cost

$$\text{Average ordering cost: } \frac{Co}{\left(\frac{Q}{D}\right)} = D \times \frac{Co}{Q}$$

Average inventory cost

$$\text{Average inventory cost: } Ch \times \frac{Q}{2}$$

Optimal Ordering Quantity: Q*



$$Q^* = \sqrt{\frac{2 \times Co \times D}{Ch}}$$

Economic Order Quantity: Quantity discounts

Solution strategy

1. Solve the problem without taking into account the discounts.
2. If Q* is large enough to obtain the largest discount then we are done.
3. If not, analyze the total cost incurred if we order the minimum size so that we obtain the discounts (analyze all discounts).
4. Select the quantity Q that gives the minimum cost.

Powered by Koffee

PBK is a new campus coffee store. PBK uses 50 bags of whole bean coffee every month, and you may assume that demand is perfectly steady throughout the year. PBK has signed a year-long contract to purchase its coffee from a local supplier, Phish Roasters, for a price of $25 per bag and an $85 fixed cost for every delivery, independent of the order size. The holding cost due to storage is $1 per bag per month. PBK managers figure their cost of capital is approximately 2 percent per month.

Questions

a. What is the optimal order size, in bags?

b. Given your answer in (a), how many times a year does PBK place orders?

c. Given your answer in (a), how many months of supply of coffee does PBK have on average?

d. On average, how many dollars per month does PBK spend to hold coffee (including cost of capital)?

Suppose that a South American import/export company has offered PBK a deal for the next year. PBK can buy a year's worth of coffee directly from South America for $20 per bag and a fixed cost for delivery of $500. Assume the estimated cost for inspection and storage is $1 per bag per month, and the cost of capital is approximately 2 percent per month.

e. Should PBK order from Phish Roasters or the South American import/export company?

Quantitatively justify your answer.

Solution Powered by Koffee

D = Demand = 50 bags/month
Price = 25$/bag
Co = Delivery Cost = $85/order
Storage cost = $1/month per bag
Capital cost = 2% per month (i.e. $25*0.02 = $0.50 per month)
Ch = $1 + $0.5 = $1.5

a) Q* = sqrt(2*Co*D/Ch) = sqrt(2*85*50/1.5) ~= **75 bags**

b) Annual demand = 12 x 50 = 600, orders per year = 600/75 ~= **8**

c) Max number of months of supply = Q/D = 75/50 = 1.5 months.
Min number of months of supply = 0. Average = **0.75 months**.

d) Avg inventory = 75/2 = 37.5.
Monthly inv cost = Ch * Q/2 = 37.5 * $1.5 = **$56.25**

Solution Powered by Koffee

e) Per month cost with local supplier = 50 * $25 + $1.5*75/2 + 85*50/75 ~= $1,362.9
Per year cost = 12* $1,362.9 = $**16,354.8**

Per year cost with South American supplier:
First we calculate unit inventory cost/year:
Capital cost cost per bag per year: $20*0.02* 12 = $0.4 * 12 = $4.8
Storage cost per bag per year: $12
Ch = $4.8 + $12 = $16.8

Total costs per year=
600*$20 + 16.8*600/2 + 500 = $**17,540**

Local supplier is preferred!

EOQ under stochastic demand

What is the objective and what are the constraints?

Objective: Minimise the annual ordering and inventory costs

Constraint: a stock-out probability threshold.

Example: We want that the probability of a stock-out per cycle is at most 5%.

Continuous Review Policy: A heuristic method

Finding the optimal solution to this problem is a complex task.

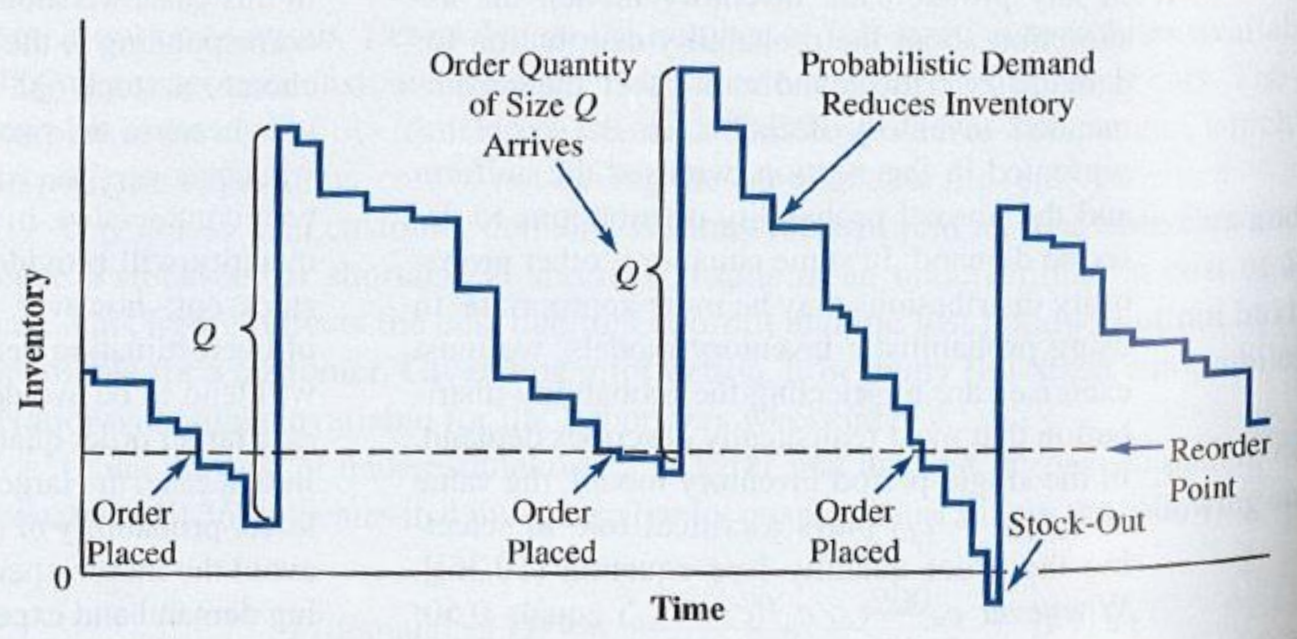
There exists an approximate method that works very well using the EOQ model.

The method consists in two steps

- Step 1: Determine how much to order (Q)
- Step 2: Find out the re-order point (r)

Continuous Review Policy

FIGURE 10.10 INVENTORY PATTERN FOR AN ORDER-QUANTITY, REORDER POINT MODEL WITH PROBABILISTIC DEMAND

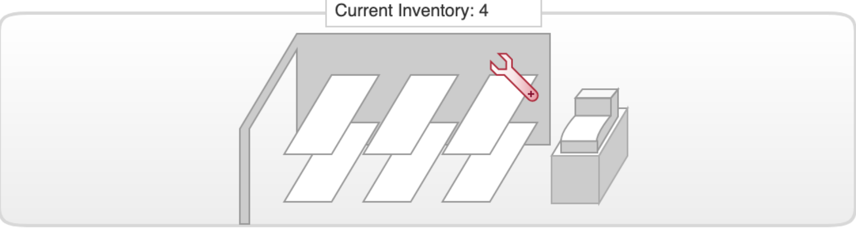


Inventory Simulation Game

Scenario 1 - Week 1

Adjustable Wrench	Holding Cost 0.04 $/(unit)(week)	Initial Mean Demand 20 units	Please place your order which will arrive in Week 2
	Ordering Cost 6.3 $/order	Last Week's Demand 21 units	Number of Units SUBMIT ORDER

Current Inventory: 4



	1	2	3	4	5	6	7	8	9	10	11	12
Inventory Level Beginning of Week	25											
Inventory Level After Unloading	25											
Customer Demand	21											
Inventory Level End of Week	4											
Quantity Ordered												
Order Costs ($)	0											
Holding Costs ($)	0.16											
Total Cost incl. Stockout Penalty ($)	**0.16**											
Total Cumulative Cost ($)	**0.16**											

Features

- Finite horizon
- Demand has a random component

Inventory Simulation Game

Question 1 ☑	If holding costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
Question 2 ☑	If ordering costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
Question 3 ☑	In the simulation, the truck delivers products the week after you submit the order. This is essentially a zero week "lead time." Assume that the lead time increased to two weeks. Would your order quantities tend to be a) larger, b) smaller, or c) the same?

Question 1

Larger

Smaller

Same

Inventory Simulation Game

Question 1 ☑	If holding costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
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Question 2

Larger

Smaller

Same

Inventory Simulation Game

Question 1	☑	If holding costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
Question 2	☑	If ordering costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
Question 3	☑	In the simulation, the truck delivers products the week after you submit the order. This is essentially a zero week "lead time." Assume that the lead time increased to two weeks. Would your order quantities tend to be a) larger, b) smaller, or c) the same?

Question 3

Larger

Smaller

Same

Inventory Simulation Game

Question 1	☑	If holding costs increased, would your order quantities tend to be a) larger, b) smaller, or c) the same?
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Question 3	☑	In the simulation, the truck delivers products the week after you submit the order. This is essentially a zero week "lead time." Assume that the lead time increased to two weeks. Would your order quantities tend to be a) larger, b) smaller, or c) the same?

Solution to question 1: If holding cost increase, the firm has the incentive to make smaller orders to reduce the average inventory. This means the firm will also make more frequent orders.

Solution to question 2: If ordering costs increase, the firm has the incentive to make larger orders so that orders are carry out less frequently.

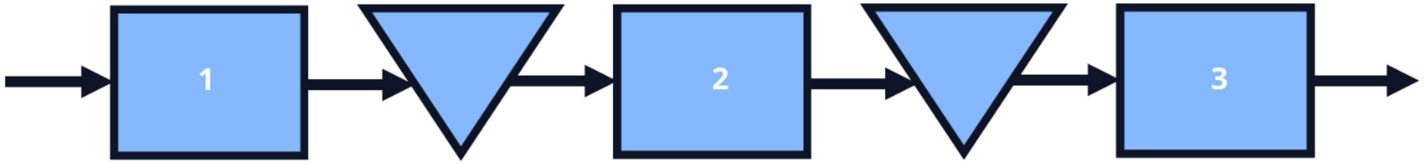
Solution to question 3: If there is no variability, the order quantity should be the same. However, when there is high demand variability, order size should increase to increase buffer against variability over a longer period.

Key Takeaways

- There are many different reasons why firms hold inventory
- Processes with setup times:
  - Capacity depends on the batch size
  - Large batch size: high inventory levels
  - Small batch size: step becomes the bottleneck and slows down the process
  - Optimal batch size: batch size needed match the desired flow rate
- Processes with setup costs: Economic order quantity (EOQ) model
  - Large orders: high inventory costs, less frequent orders
  - Small orders: low inventory costs, more frequent orders
  - Optimal EOQ: order size that minimises the sum of the inventory costs and the ordering costs

Practice question 3 (past challenge)

Consider the following process that is composed of three steps in the following sequence:



Step 1: Activity time = 70 seconds per part. Setup time = 0.5 hours.

Step 2: Activity time = 50 seconds per part. Setup time = 0.49 hours.

Step 3: Activity time = 60 seconds per part. Setup time = 2 hours.

Work is processed in batches at each step. Before the batch is processed at a step, the machine must be set up.

What is the capacity of Step 1 if the batch size is 90?

*State your answer in **number** of units per hour.*

Solution to practice question 3

$$\text{Capacity of step 1: } \frac{90}{\left(0.5+\frac{70}{3600}\times 90\right)} = 40 \text{ units per hour}$$

Practice question 4 (past challenge)

Question:

If the batch size is sufficiently large, what step is the bottleneck?

Solution to practice question 4

When the batch size is sufficiently large, the setup time is not relevant to find out the bottleneck step. This is because the setup time represents almost 0 per cent of the total time of a batch cycle. In this scenario, the task with the highest processing time is the one that is the bottleneck step. Thus, the bottleneck step is Step 1 as this is the step that has the highest processing time.

Problem with setup times

Q6.1* **(Window Boxes)** Metal window boxes are manufactured in two process steps: stamping and assembly. Each window box is made up of three pieces: a base (one part A) and two sides (two part Bs).

The parts are fabricated by a single stamping machine that requires a setup time of 120 minutes whenever switching between the two part types. Once the machine is set up, the activity time for each part A is one minute while the activity time for each part B is only 30 seconds.

Currently, the stamping machine rotates its production between one batch of 360 for part A and one batch of 720 for part B. Completed parts move from the stamping machine to the assembly only after the entire batch is complete.

At assembly, parts are assembled manually to form the finished product. One base (part A) and two sides (two part Bs), as well as a number of small purchased components, are required for each unit of final product. Each product requires 27 minutes of labor time to assemble. There are currently 12 workers in assembly. There is sufficient demand to sell every box the system can make.

a. What is the capacity of the stamping machine?

b. What batch size would you recommend for the process?

Solution

- Capacity at assembly can be found from the fact that 12 units come out every 27 minutes.

- The capacity at the stamping machine is determined by the batch size. Larger the batch size, larger the capacity.

- Capacity of stamping machine for a given batch size B
  - = B/ (240 minutes + 2 min/unit * B)

  We incur 240 minutes of setup and the actual activity time for a unit is 2 minutes.
  Currently B = 360, so the capacity at the stamping machine is 360/(240+2*360) = 0.375 units per min

Solution

The desired flow rate is 12 units / 27 min ~= 0.444 units per minute.

The capacity at the stamping machine must match the desired flow rate.

Capacity at stamping machine = B/(240+2B) = 12/27 = desired flow rate.

27 B = 240 x 12 + 24 B

3 B = 240 x 12 and

Recommended batch size = B = 960 units

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